Observer Patch Holography
A Compact Case for an Observer-First Unification Program

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Abstract

Observer Patch Holography starts with finite self-reading systems that expose boundaries, keep records, compare overlaps, and repair disagreement. The same architecture supports exact public-record and event-algebra results, Lorentz kinematics on a spherical screen, a conditional Einstein composition, and a theorem forcing the complete twelve-port response to have the Standard Model Lie type, without choosing an ambient gauge group. Deterministic simulations measure a Lorentzian event form with one time and three space directions across a support-adjusted three-rung path. Interval arithmetic certifies unique roots of the declared pixel maps. A positive-chamber circulant theorem gives the Koide relation exactly, while finite capacity identities supply a conditional de Sitter time-advance mechanism. A target-clean finite source also inhabits exact causal, seam-transport, and local operator domains, whose boundaries stop short of a continuum Lorentzian manifold or a physical clock. The physical attachments are explicit. Precommitted comparisons use custody-bound target definitions, kill bands, precision floors, and decision rules. A sorry-free Lean library, exact code receipts, and reproducible simulations make the case testable at each step.

Observer Patch Holography (OPH) asks whether physics can be reconstructed from the minimum machinery required for a public world. A patch has local state, a boundary, records, readback, and repair moves. Each patch sees a fragment. Agreement across shared boundaries turns private descriptions into public facts. The selected world is a stable normal form,

$$T(\mathfrak{U}) = \mathfrak{U}.$$

The theory rests on three core axioms.

The three axioms

1. Oriented twelve-port observer screen. There exists an observer patch net on an oriented spherical screen: at every finite resolution each local carrier has twelve primitive boundary ports forming the vertices of an oriented triangular boundary with 30 edges and 20 faces, combinatorially the boundary of an icosahedron; carriers join through typed seams and coherent triple overlaps, refine to an oriented spherical support, and expose local state, readback, records, repair moves, and checkpoints. Formally: for every regulator r there is a typed object $\mathfrak{N}_r = (\mathcal{P}_r, \mathcal{A}_r, \mathcal{R}_r, \mathcal{I}_r, \mathcal{U}_r, \mathcal{C}_r, N_r, S_r, b_r)$ whose carriers carry twelve primitive central port projections and the exact boundary packet $K = (P, E, F, o)$, joined by seam algebras with coherent triple-overlap cocycles into a nerve N_r with a degree-one bridge b_r to the oriented spherical support S_r, all commuting with refinement. A complete carrier has a finite-dimensional unitary reversible response space and an injective real-linear port derivative $D_r : \mathbb{R}^{P_r} \to \mathfrak{u}(H_r)$ with positive trace pairing, commutator-closed image, and refinement-natural quotient-visible response.

2. Observer agreement. Observers operating on the screen agree on the meaning of the data they jointly interpret. Formally: the interpretation map $\mathcal{J}_r : \mathsf{Data}_r \to \mathsf{Meaning}_r$ is natural with respect to every visible overlap restriction, recharting, seam translation, higher-overlap map, federation map, and refinement map on accepted public data. Accepted reversible overlaps form a groupoid surjecting onto the proper carrier automorphisms. For each such automorphism, one closed path has a projective implementer whose adjoint action is covariant on D_r and lies in the response-generated identity component modulo the pointwise centralizer.

3. Conditional maximum randomness. Everything that observer agreement leaves unconstrained is maximally random. Formally: the realized state is the information projection of an exact reference family onto the convex set of compatible local state families satisfying the finite observer-visible constraints. The finite A1-generated observer cover is state-determining on that feasible set, and its exact weights are strictly positive: $\rho_r = \arg\min_{\rho \in \mathcal{K}_r} \sum_P w_{r,P} D(\rho_{r,P} \| \tau_{r,P})$.

The first axiom constrains architecture, the second meaning, the third state selection. None of them contains a gauge group, a particle list, a recovery law, or a rule that fixes field content or multiplicity. Every result below states its axiom dependencies, further premises, mathematical domain, and physical boundary. The complete plain and formal statements, scope clauses, and caveats are collected in the canonical axiom reference.

One modeling principle sits beside the axioms without joining them. The framework models the universe as a self-referential fixed point: the simulating and the simulated description are one system. Every quantity with a construction-side and a readback-side reading therefore obeys a consistency requirement, that its readback return map close on a unique fixed point on the declared domain, and an identity requirement, that the two readings name one invariant, with each identity a separately typed physical attachment. This principle is what imposes the two quantitative closures: the screen-grain equation $P = \varphi + \sqrt{\pi}/A_T(P)$, whose interval-certified fixed point stands at the few-parts-per-million level against the measured fine-structure constant as a source-map result with its hadronic transport open, and the capacity candidate $\ln(N/\pi) = 6\pi/(P\,\alpha_U) - P/24$, which corrects the electroweak bridge by the shared-cut $\mathbb{Z}_6$ protected-reserve mean, carries four named premises with a pending horizon-record attachment, agrees with the measured cosmological constant retrospectively, and is rejected outright if the measured capacity ever exceeds the uncorrected bridge value.

The proposal earns attention because several familiar structures arise from this small basis. They also arrive with different evidence types. Exact theorems, interval certificates, conditional physical compositions, and simulation measurements occupy separate evidence classes throughout this paper.

The claim in one sentence

Finite observer-like patches provide a common construction for public quantum records, Lorentz kinematics, a conditional Einstein branch, compact gauge structure, matter representations, and quantitative closure tests. The mathematics fixes a substantial skeleton. Physical carrier identifications determine how much of that skeleton describes our universe.

1. Ten receipts that carry the case

The following count contains the strongest independent pieces of evidence. Every row states its boundary in the same breath as its result.

	Receipt	Result and boundary	Evidence
1	Public records and finite event algebra	Terminating repair gives protected normal forms on the declared confluent carrier class. The completed central record surface obeys Born probabilities, Lüders conditioning, and the Tsirelson bound. On the twelve-port carrier, atomic signed record events generate integer loads and conservative repairs terminate by exact quadratic descent.	Lean and exact finite code [1,2,10]
2	Lorentz kinematics and three observer-frame dimensions	On the certified spherical support branch, $\mathrm{Conf}^+(S^2) \cong \mathrm{SO}^+(3,1)$ and $\dim[\mathrm{SO}^+(3,1)/\mathrm{SO}(3)] = 3$. Event population is a separate physical receipt.	Theorem chain [1,3]
3	Einstein composition and a finite signature test	The typed modular, null-stress, entropy, and small-ball premises compose to the Einstein relation. On a declared chart with one ancestry coordinate and three spectral coordinates, a support-adjusted three-rung path measures inertia $(1,3)$ with cone margins $-5.62, -3.22, -1.41$. A same-size support-width control changes the inertia to $(2,2)$. The instrument tests signature and does not select the chart dimension. One target-clean finite source also inhabits exact causal, atlas, seam-transport, and local operator domains. Its six local fits include one $(0,4)$ chart, and its cone margins are negative.	Lean implication, instruments, simulation [3,11]
4	Standard Model local gauge algebra without a chosen gauge group	The complete reversible port response in A1 and endogenous overlap transport in A2 imply an injective compact commutator-closed twelve-dimensional current with inner A_5 action. Its one-dimensional fixed space and compact classification force $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$. No ambient Lie group, factor dimension, or matrix current is inserted. The finite-source simulator derives the carrier module and $R = -J$. Ordered current tomography and same-current holonomy are work in progress.	Lean, compact classification, exact controls [3,10]
5	Global quotient and one-generation representation witness	For the declared tensor representation, trace balance gives $S(U(3) \times U(2))$, the common action kernel is $\mathbb{Z}_6$, and an exterior algebra branches into one fifteen-state Standard Model generation with the anomaly arithmetic. The physical global quotient requires a source-derived complete character category and a same-source loop-to-kernel identity. Among the single complete faithful screen bands in $3 \leq N_g \leq 5$, the exact Laplacian cost selects rank three, and the declared unitary simulator recovers that band at its lowest positive generator frequency. Tensoring the band with the declared generation table gives a conditional complex rank-45 candidate. The chirality grading and diagonal $\mathbb{Z}_6$ action belong to that table. A separate local-domain receipt checks the declared extension $D_\sigma \otimes I_{45}$ and conditional inheritance of its positive finite gap. The source does not select this matter action. The twelve-port Spin packet and the 8,662-node local domain have no certified source, domain, or transport bridge. Matter-pole identification, continuum Spin/locality, physical seam selection, and laboratory attachment are open.	Lean and exact code [3,10]
6	Pixel fixed-point arithmetic	Outward-rounded interval certificates prove existence and uniqueness for each declared pixel map and exclude a second root across its analytic domain. The fine-structure comparison is diagnostic because the physical transport map is incomplete.	Interval arithmetic [5,10]
7	Charged-lepton interval diagnostic	The measured electron, muon, and tau masses lie inside outward-rounded transport enclosures. Their logarithmic half-width is 1.732%, giving one-sided multiplicative widths of -1.72% and $+1.75\%$. Target-anchored ratios and empirical transport make this a closure diagnostic.	Interval certificate [6,10]
8	Positive-chamber Koide theorem	A Hermitian three-cycle response obeys $Q = \frac{1}{3} + \frac{2}{3}(\lvert b\rvert/a)^2$. Hence $Q = 2/3$ exactly at $\lvert b\rvert/a = 1/\sqrt{2}$. Equal finite event blocks supply this balance under the declared tracial packet. Under the balance and ordering premises the measured electron and muon masses fix the tau mass inside a 72-eV enclosure, 0.43σ from measurement. The conditional test has a fixed precision floor and rejection rule. The phase and physical family attachment are work in progress.	Lean and exact code [7,9,13]
9	Finite de Sitter identities	Pure de Sitter normalization, the finite entropy maximum, the exact capacity transfer law, its analytic curvature, and the port/edge line-graph identity are exact. The gravitational shock sign requires the stated horizon and observer dictionaries.	Lean and exact code [8,10]
10	Precommitted comparison tests	Comparison contracts use custody-bound target definitions, kill bands, precision floors, and decision rules. The frozen registry records each target and its ancestry. A target without a custody-bound source packet carries no predictive weight. The coefficient-free quartic entropy lepton-shape candidate is excluded at 16.7% by its own exact golden-ratio gap theorem.	Custody records [13]

The accompanying Lean library contains more than 1000 theorem and lemma declarations. The count includes positive results, premise boundaries, and countermodels. It is useful because the formal statements expose assumptions that prose can otherwise hide. The finite receipts bind inputs, outputs, tolerances, and negative controls to reproducible artifacts [9].

2. Public reality and quantum records

Take a finite family of patches x_i, each with an observable boundary map. For every overlap $e = (i, j)$, the two induced records must agree. A repair move changes local state while preserving protected readout. On a terminating, frustration-free carrier with a confluent accepted repair relation, every schedule reaches the same quotient normal form. Exact code exhausts all six declared overlap schedules on the reference carrier and returns the same normal form [2,10]. The general confluence hypothesis is part of the carrier contract. Arbitrary local repair does not imply it.

Once a compare, write, and verify slice has completed, its accessible events generate a finite central record algebra. If P_E is the projector for an event E and ρ is the normalized state, then

$$\Pr(E) = \mathrm{Tr}(\rho P_E), \qquad \rho \mid E = \frac{P_E \rho P_E}{\mathrm{Tr}(\rho P_E)}.$$

The first expression is the Born probability on this finite surface. The second is Lüders conditioning. The corresponding Clauser–Horne–Shimony–Holt (CHSH) parameter satisfies

$$|S_{\mathrm{CHSH}}| \leq 2\sqrt{2}.$$

These identities belong to a 112-declaration audited finite event-algebra development [1,10].

The result gives quantum record arithmetic from public observability. It does not construct an arbitrary continuum quantum field theory. Its value is more specific: a finite observer contract produces the probability and update rules required by physical branches built on it.

3. Lorentz kinematics and the Einstein branch

An oriented conformal spherical support has the connected symmetry

$$\mathrm{Conf}^+(S^2) \cong \mathrm{PSL}(2, \mathbb{C}) \cong \mathrm{SO}^+(3, 1).$$

The space of unit timelike observer frames is

$$H^3 \cong \frac{\mathrm{SO}^+(3,1)}{\mathrm{SO}(3)}, \qquad \dim H^3 = 6 - 3 = 3.$$

This is an exact kinematic result on the spherical support branch. Local icosahedral carrier geometry, the federation nerve, and the global S^2 support are distinct objects. A populated four-dimensional event base requires record separation, local charts, an open-image condition, affine transition data, a quadratic cone, and causal reachability [3].

On one common algebra-state tower, normalized modular flow supplies a local clock, half-sided modular inclusions supply positive null translations, null tomography assembles local stress, and generalized-entropy stationarity with small-ball geometry gives

$$G_{ab} + \Lambda g_{ab} = 8\pi G_{\mathrm{geom}} \langle T_{ab} \rangle.$$

The Lean development proves the typed algebraic composition. A source-derived physical tower, the continuum limit, and the physical identification maps are premises [3,10].

The simulation branch fits an event form to coordinates and causal labels produced by repair dynamics; its coefficients are not inserted into the readout. The selected path uses $(16{,}384, 128, 96)$, $(65{,}536, 256, 96)$, and $(262{,}144, 512, 384)$ for carrier count, observer count, and support width. The held-out event form has inertia $(1, 3)$ at every selected rung. Its cone margin moves through

$$-5.6, \qquad -3.2, \qquad -1.4.$$

Coupling spread decreases beside it. At 262,144 carriers, the support-width 96 control has 312 cross-observer edges and inertia $(2, 2)$; the support-width 384 row has 1,062 edges and inertia $(1, 3)$. Configurations, primary arrays, hashes, and regeneration scripts are public [10]. The data establish a reproducible support and cross-read sensitivity on the instrumented branch. They do not establish an invariant-density convergence law, an infinite-scale limit, or the missing cap-state thermal receipt.

A finite source inhabits the local operator domain

One deterministic target-clean capture at 16,384 carriers supplies a finite causal complex with 2,304 events and exact acyclicity. Its declared chart combines one ancestry-depth coordinate with three spectral embedding columns. The four columns are nondegenerate on the sample, and the global held-out form has inertia $(1, 3)$. Six closed observer-visibility neighborhoods have exact induced affine wiring transitions, cocycles, orientation, and time orientation. Five local quadratic fits have inertia $(1, 3)$; one has $(0, 4)$, and the measured cone margins are negative.

On its observer-visible seam complex, 8,662 carriers and 11,816 seams contain 38 triangles. The declared reversing seam transport frustrates all 38. Two coordinate routes through the same $\mathbb{F}_2$ rank implementation agree with the exact rank identity and give lift-ambiguity rank 3,117. Scalar, chiral, and gauge sections carry a declared unit-counting measure and a sign-twisted local derivative. Exact integer evaluations on deterministic sections test its adjoint, kinetic, covariance, gluing, refinement, and boundary relations. The signed-graph rank theorem gives a zero twisted kernel. One target-clean provenance graph binds the construction, and an isolated rerun of the same producers reproduces the canonical receipt content [10].

This finite result gives the particle, clock, and defect calculations a common typed domain. It does not supply open manifold charts, a positive cone margin, continuum Stiefel–Whitney identification, physical field selection, or a continuum operator limit.

Held-out measurement and control

The observed signature is held out from the update rule, persists across three named configurations, changes under the same-size support-width control, and sits inside a typed theorem chain that states every physical premise. This combines an exact implication with a direct computational measurement.

4. Why the complete twelve-port response forces the Standard Model gauge algebra

The declared carrier lineage realizes the twelve ports with equal algebra traces, a realization property rather than axiom content, with oriented icosahedral incidence. Its proper rotation group is the alternating group A_5. Exact finite calculations give the distance profile, the antipodal pairing, the sixty proper rotations, the rank-three Gram frame, and the decomposition of the real port-coefficient space

$$P_{12} \cong_{A_5} \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}.$$

Complete reversible port response in A1 supplies an injective commutator-closed map $D : P_{12} \to \mathfrak{u}(H)$. Endogenous overlap transport in A2 implements every proper carrier recharting inside the response-generated group. The image is a compact twelve-dimensional Lie algebra with a one-dimensional A_5-fixed space. Compact classification then gives

$$\operatorname{im} D \cong \mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3).$$

The centreless alternative is $\mathfrak{su}(2)^4$. An inner A_5 action preserves every simple factor. Since A_5 is simple, each restricted action is trivial or faithful and has fixed dimension three or zero. The total cannot equal one. The surviving centre has dimension one and the semisimple part has dimensions $3 + 8$ [3,10].

Gauge structure without a chosen gauge group

The argument starts with the operational response of the twelve ports. A1 makes that response faithful, complete, unitary, and closed under sequential commutators. A2 makes every proper carrier recharting internal to the same response. Incidence supplies the A_5-module and its single fixed line, but it leaves many equivariant response laws. The response and agreement clauses remove that freedom together. The fixed-line obstruction and compact classification leave the local Standard Model gauge Lie algebra as the unique result. No ambient compact group, factor dimension, matrix current, charge table, or particle assignment is a premise. The theorem fixes the abstract local Lie algebra. Source realization, matter action, coupling normalization, the global quotient, and laboratory attachment remain separate. A descent repair law is required for branches that consume settled public records; it is not an extra premise of this Lie-algebra theorem.

Target-blind port impulses and local recurrence readback derive $R = -J$ and its four band signs. The charged-double-triplet matrices give an exact conditional realization

$$\mathfrak{u}(3) \oplus \mathfrak{so}(3) \cong \mathfrak{u}(1) \oplus \mathfrak{su}(3) \oplus \mathfrak{su}(2).$$

The source contains no ordered reversible current tomography or closed overlap words tied to that same current. Matrix realization, coupling normalization, laboratory attachment, and identity with the independently reconstructed categorical current are work in progress.

Trace balancing extends the same block construction to $\mathfrak{s}(\mathfrak{u}(3) \oplus \mathfrak{u}(2))$. Its connected group is

$$S(U(3) \times U(2)) \cong \frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}.$$

The six-element kernel is exact for the declared tensor table. The independent axis calculation has Smith invariants $(1, 1, 1, 1, 1, 6)$ after diagonal and zero-sum axis relations are declared. The carrier measures the six axes and their deck action. It does not select those relations. Equality of the two group orders and an isomorphism between chosen generators therefore give a conditional algebraic match, not a physical global-form selection. A source-derived complete character category, a same-source loop-to-kernel identity, four-dimensional instanton normalization, and laboratory flux attachment are work in progress.

The trace-balanced block $V = \mathbb{C}^3 \oplus \mathbb{C}^2$, with hypercharges $(-1/3, 1/2)$, has the exterior branching

$$\Lambda^2 V \oplus \Lambda^4 V = Q \oplus u^c \oplus e^c \oplus d^c \oplus L.$$

It contains fifteen states, the Standard Model hypercharges for one generation, the expected three one-Higgs invariant lines, cancelled gauge and mixed anomalies, and exactly four weak doublets. This is a precise representation witness. The finite response source does not select the charged-double-triplet current or this matter action. Laboratory matter identification, exclusion of extra light sectors, and attachment of three physical families require separate source receipts [3].

The screen coefficient module also gives an exact family-band selection under its named premises. Its four bands have ranks $1, 3, 3, 5$. Restricting to single complete faithful objects in the conditional window $3 \le N_g \le 5$, the Laplacian costs $5-\sqrt{5}, 6, 5+\sqrt{5}$ select the first rank-three band uniquely. The declared unitary simulator recovers that projector at its lowest positive generator frequency. Identifying this finite response mode with physical matter families requires matter-pole identification, continuum Spin/locality, physical seam selection, persistence, and laboratory receipts. Tensoring the rank-three band with the declared generation table gives a conditional rank-45 candidate whose chirality and diagonal $\mathbb{Z}_6$ data come from that table. A different receipt checks the declared local extension $D_\sigma \otimes I_{45}$ and conditional inheritance of the signed seam operator's positive finite-domain gap. This action is not source-selected. The twelve-port Spin packet has no certified bridge to the 8,662-node local domain. The finite gap is distinct from the compact-gauge repair spectrum and the continuum Yang–Mills mass gap.

The record-counting mechanism is operational on the same twelve-port architecture. A write appends $+1$, a retraction appends -1, and the readback sums these atomic events at each port. A conservative repair transfers one whole unit across a seam whenever its oriented mismatch has magnitude at least two. If $V(N) = \sum_i N_i^2$, a repair from the larger endpoint to the smaller one changes V by $-2(d-1)$, where $d \ge 2$ is the mismatch. The repair therefore terminates. Divisibility of the total load by twelve is a necessary consensus condition. An explicit eighteen-move witness settles the declared full-pile packet. The minimum admissible move count is natural under all sixty carrier rotations and the declared refinements. A half-unit display rescales the same event graph and threshold, so it changes units rather than the atomic mechanism or its cost.

The finite theorem forces the local Standard Model gauge Lie algebra. The separate transportable-sector route reconstructs a compact group, and the declared Standard Model sector packet has the same local type. A theorem identifying the two current objects on one physical source is a separate obligation.

5. Arithmetic closure and charged-lepton diagnostics

The local pixel fixed point

The local screen cell is asked to reproduce its readable electromagnetic width:

$$P = \varphi + \frac{\sqrt{\pi}}{A_T(P)}, \qquad \varphi = \frac{1+\sqrt{5}}{2}.$$

Here $A_T(P)$ is the inverse electromagnetic coupling emitted by a trial cell through the declared Thomson-limit map. For each declared map, outward-rounded interval Banach certificates prove existence and local uniqueness. A 256-piece derivative enclosure proves global at-most-one across the analytic domain, with an empty exceptional set [5,10]. The gauge-width map has the certified root

$$\alpha^{-1} = 137.035660136946577\ldots$$

beside the 2022 Committee on Data of the International Science Council (CODATA) value 137.035999177(21) [11]. The relative separation is 2.5×10^{-6}.

Neither source fixed-point solve imports the measured Thomson value. Their physical interpretation requires the missing source-derived, same-scheme hadronic transport. A separate mixed diagnostic combines the source root with a coupling evaluated at the CODATA-derived comparison pixel. That quantity is a comparison, not a source-only prediction. The certified roots do not constitute an independent determination of the physical fine-structure constant.

The charged-lepton interval surface

The charged-lepton calculation combines a finite eight-path carrier with an empirical electromagnetic transport packet. Independent 100-digit decimal recomputation gives outward-rounded enclosures with logarithmic half-width 1.732%, corresponding to one-sided multiplicative widths of −1.72% and +1.75%. The measured triple lies inside all three enclosures. A narrower conditional eight-register coordinate lies about 84 parts per million from the comparison triple [6,10].

Measured mass ratios and transport anchors enter this branch, so these numbers are diagnostics of internal closure. They give a sharp test surface without claiming a source-only mass prediction. A physical charged determinant line, an absolute clock, and a source-emitted transport bridge are required for an endpoint prediction.

An exact positive-chamber Koide theorem

Let $R^3 = I$ and consider the Hermitian three-cycle response

$$C = aI + bR + \bar{b}R^2.$$

When its three eigenvalues are nonnegative and are read as square-root masses, the normalized mass coordinate satisfies

$$Q = \frac{\sum_i m_i}{\left(\sum_i \sqrt{m_i}\right)^2} = \frac{1}{3} + \frac{2}{3}\left(\frac{|b|}{a}\right)^2.$$

Consequently,

$$Q = \frac{2}{3} \qquad \Longleftrightarrow \qquad \frac{|b|}{a} = \frac{1}{\sqrt{2}}.$$

The phase cancels from Q and jointly controls the two mass ratios. Equal rank-two event blocks in the finite tracial Gelfand–Naimark–Segal packet give the required modulus balance exactly [7,10]. Lean checks the circulant identity and equivalence. Physical chiral-family attachment and phase selection are work in progress, so the theorem is independent of the charged-lepton diagnostic above.

Under the balance and mass-ordering premises, the measured electron and muon masses determine the tau mass through one quadratic. The outward-rounded enclosure is [1776.968991, 1776.969063] MeV, a 72-eV window sitting 0.43σ from the measured 1776.93 ± 0.09 MeV, with the spurious root excluded below the muon mass and the premise ancestry declared: the balance premise is target-informed by the measured triple, so this is a conditional postdiction. The registered test has a precision floor of 0.045 MeV. A qualifying average outside the 72-eV window refutes the balanced-circulant premise; one inside passes its registered test. The target definition, kill band, decision rule, and calendar custody are fixed [13].

The lepton shape carries a sharpened boundary of its own. The stabilizer theorem leaves two projective shape parameters, so symmetry geometry alone predicts nothing, and the first coefficient-free candidate potential, the quartic truncation of the entropy expansion on the quintet carrier, is excluded: its only simple-spectrum orbit has sorted-gap ratio exactly the golden ratio, 16.7% from the observed centered log-mass ratio in the closest orientation. The exclusion is itself an exact theorem, with the orbit eigenvalues $(-5/2,\ 5/4 \mp (3/4)\sqrt{5})$ in closed form [6,9]. The lepton-shape object requires a source mechanism beyond that truncation.

6. Fixed capacity and the de Sitter shock sign

For pure de Sitter space of radius L in spacetime dimension d,

$$r_c = L, \qquad \kappa = L^{-1}, \qquad \mu^2 = (d-2)\kappa r_c = d-2 = \lambda_{\ell=1}(S^{d-2}).$$

The radius and cosmological constant cancel. The smooth $\ell = 1$ modes are generated by de Sitter isometries and form the gauge kernel of the imported shock operator [8,13].

For finite sector dimensions d_i, total dimension $M = \sum_i d_i$, and sector probabilities p_i,

$$S_{\mathrm{gen}}(p,d) = -\sum_i p_i \log p_i + \sum_i p_i \log d_i = \log M - D_{\mathrm{rel}}(p \| d/M),$$

where D_{rel} is relative entropy. Thus $p_i = d_i/M$ gives the exact maximum $S_{\mathrm{gen}}^{\max} = \log M$. If an observer receives a fraction f of a fixed horizon-observer capacity and the horizon sectors deplete uniformly, every admissible integer transfer obeys

$$\Delta S_{\mathrm{gen}}^{\max} = \log(1-f) < 0.$$

The positive-real interpolation is strictly decreasing and concave. Its zero-transfer point is a one-sided boundary maximum. The associated logarithmic sector observable has symmetric-point Hessian

$$\operatorname{Hess}\mathcal{A} = \frac{1}{nd^2}\left(I - \frac{2}{n}J\right).$$

Its fixed-total tangent curvature is positive, its homogeneous curvature is negative, and its symmetric-point gradient is nonzero. The one-sided capacity transfer carries the sign; an interior fixed-capacity extremum is excluded [8,10].

On the twelve-port icosahedral graph, the raw Laplacian spectrum and regular line-graph identity are exact. If the rotation triplet is an exact gauge sector and the physical shock kinetic term is the scaled nearest-neighbour port or edge Laplacian, normalization at that triplet gives

$$\{-2,\ 0,\ 1+3/\sqrt{5},\ 1+\sqrt{5}\}.$$

The edge carrier adds only eigenvalue 10 with multiplicity 18. Identifying capacity with horizon area, the transfer coordinate with observer mass, and the finite operator with the gravitational shock supplies a conditional time-advance mechanism. The finite identities make none of those physical identifications. They supply no positive cyclic static-patch trace. The obstruction of Chen, Stanford, Tang, and Yang excludes that trace proposal rather than the finite identities above [12].

7. Why the cumulative case is worth testing

Each receipt has a separate mathematical domain. OPH becomes interesting because the receipts share one finite architecture. Public records lead to the event algebra. Spherical support leads to Lorentz kinematics. Modular and entropy data lead to the Einstein composition. Icosahedral carrier data lead to the gauge coefficient algebra and an exact matter witness. The same finite readback idea produces the pixel closure, the tracial Koide balance, and the capacity transfer law.

The dependencies are visible enough to be attacked. A proposed carrier must emit local state, ports or boundaries, readback, records, repair moves, and a public evidence bundle. Physical claims require source-bound maps between the finite objects and their spacetime, current, matter, clock, or horizon interpretations. Lean checks symbolic implications. Exact programs check finite searches and interval arithmetic. Simulations measure branches whose continuum construction lies beyond the finite theorem.

The falsification instrument uses target definitions, premise ancestry, cryptographic custody, kill bands, precision floors, and decision rules fixed before comparison [13]. The tau window has a custody-bound precision floor and decision rule. The Higgs envelope is a target-conditioned comparison without a predictive verdict. The charged-lepton anchor-gap band is a confirm-or-refute condition for a source-emitted bridge value, whose derivation is work in progress. Landing inside confirms the decomposition; landing outside refutes it.

Conditional hypotheses that fail their predeclared comparison rules are excluded from the evidentiary case in this paper. The rejected hypotheses contribute nothing to the exact record, geometry, gauge, Koide, or capacity results [13].

The program therefore has a clear scientific shape. It contains exact mathematics with recognizable physical structure, a measured finite Lorentzian-signature signal with mechanism controls, certified arithmetic closures, and explicit physical attachments that can succeed or fail. That combination supports a serious observer-first route toward unification. It does not amount to a completed physical theory of everything.

Assessment

OPH has a machine-checked core, exact finite constructions that land on the algebraic skeleton of known physics, and a reproducible support-sensitive Lorentzian-signature signal on a fixed chart. Its strongest physical conclusions depend on named carrier maps. The combination makes OPH a promising observer-first research program rather than a completed physical theory.

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